

An Augmented Reality Exergame for Improving Ski Posture through Gamified Feedback

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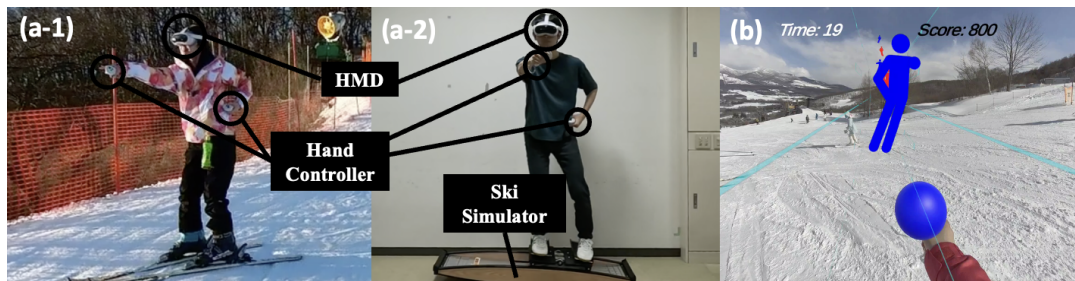


Figure 1: Overview of the ski exergame in real-world (a-1) and on a ski simulator (a-2). User's view is shown in (b).

Abstract

In this paper, we present an AR exergame that integrates ski posture training with gamification. Skiing poses unique challenges for posture correction due to its dynamic and outdoor nature, making traditional feedback methods like mirrors or video analysis impractical. Our system displays visual pose cues in AR, synchronized with skier motion, and scores the skier's accuracy in real-time. It supports use both on real ski slopes and indoor simulators using a standalone HMD. Preliminary indoor and slope tests show the feasibility of using AR for engaging and effective ski training.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction paradigms*; Mixed / augmented reality;

Keywords

Augmented Reality, Skiing, Gamification, Skill Acquisition

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1 Introduction

Alpine skiing is a competitive sport in which athletes aim to minimize descent time while executing precise turns through gates. Maintaining proper posture (a stable spine angle and hip alignment) is essential for effective skiing¹. However, novice and intermediate skiers often struggle to maintain posture, tending to lean inward. Traditional feedback methods such as mirrors or video are impractical, as skiers are continuously moving downhill.

Previous research has proposed sensor-based approaches for skiing analysis [Hasegawa et al. 2012; Kos and Umek 2018] and XR systems for feedback [Fan et al. 2016; Matsumoto et al. 2024; Wu et al. 2019]. These approaches rely on users consciously interpreting feedback; however, reproducing complex full-body actions such as ski turns remains challenging. In practice, training often involves repeating specific poses during turns (e.g., extending arms forward to align hips, Figure 1(a)), which may become monotonous and reduce motivation.

To address these limitations, we present an AR ski exergame that combines posture-corrective training with gamification. The

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¹<https://ltad.alpinecanada.org/stages/ski-for-life/technique-and-tactics>

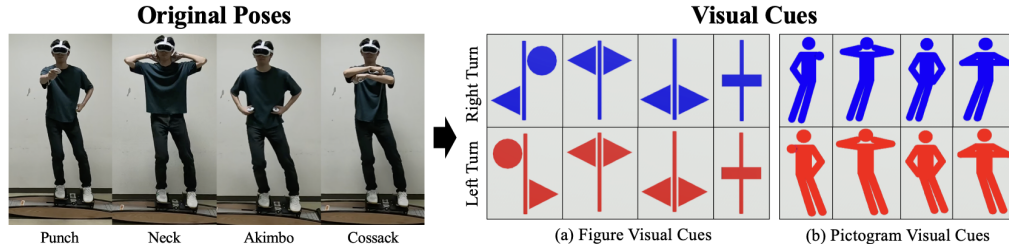


Figure 2: The list of poses and visual cues used in our system.

system displays AR visual cues for specific poses, which users perform during turns. To increase engagement, cues are shown in a randomized sequence, and users' performance is scored. This approach aims to improve posture and sustain motivation. The system operates both on slopes and with indoor ski simulators. In this paper, we describe the system and report preliminary slope experiments, followed by a discussion of its feasibility.

2 System & Implementation

2.1 Posture Correction

We propose an AR exergame that supports posture correction during ski turns while enhancing motivation. The system integrates traditional training methods, where skiers adopt corrective poses during turns. Each of the four poses is linked to a unique AR visual cue, with users synchronizing their turns accordingly. The cues are symmetrically designed to accommodate both left and right turns.

Figure 2 shows the visual cues. Two types are available: figure shapes that clearly distinguish forms and human-shaped pictograms that intuitively convey poses. Users can select their preferred cue type before training.

2.2 AR Gamification System

Figure 1(b) shows the display. Visual cues appear from the background and alternate between left and right. The user performs turns such that the HMD trajectory passes outside each cue. The cue types (Figure 2) appear in random order to maintain engagement.

When passing a cue, the system evaluates pose accuracy and provides feedback through a score. The evaluation involves: (1) recording controller positions relative to the HMD while stationary, (2) computing the distance to in-turn positions, (3) checking whether it falls below a predefined threshold, and (4) calculating a score. Feedback is provided through an AR score display and audio.

2.3 Hardware Implementation

The system operates both on slopes and indoors. To avoid connection issues reported in prior work [Colley et al. 2015], it is implemented entirely on a standalone HMD (Figure 1(a)), which leverages the pass-through function for real-world awareness. For indoor use, a ski simulator reproduces turning motions.

3 Preliminary Study on the Simulator

Prior to slope evaluation, we conducted a preliminary indoor study to assess system effectiveness and identify requirements for slope use. The setup consisted of a ski simulator (President Lux Model by

SkyTechSport²) and a PICO4 HMD. Four skiers participated with figure-shaped visual cues (Figure 2(a)) and provided feedback.

All participants completed the training, and comments such as "I tried to improve my score" (P1, P3) suggested that the scoring system enhanced engagement. However, some reported difficulty coordinating turns with the required poses, noting "It was challenging to execute turns while considering the pose" (P2, P3). Since skiing on real slopes also requires terrain and obstacle recognition, an even greater cognitive load is expected.

To address this, we used human-shaped pictogram cues (Figure 2(b)) in the slope study, as they are more intuitive and may reduce cognitive load.

4 Preliminary Study on the Slope

To evaluate the feasibility of our system on ski slopes, we conducted a preliminary study using the setup described in Section 2.3. We employed the PICO4 Ultra as the HMD, which offers a clearer view of the environment than the PICO4 used in Section 3. Four skiers tested the system and provided feedback (Figure 1(b)).

Participants can perceive both the environment and AR cues, and successfully performed ski turns while adopting the required poses. Controller tracking was reliable. These results suggest that our system is feasible for use on real slopes.

Several challenges remain to be addressed. Visual cues must align with slope contours and width, as skiers change direction while descending. Strong sunlight may cause tracking errors in infrared systems [Matsumoto et al. 2024]. Although the 105° FoV was not a major concern, some participants noted that focusing on cues reduced awareness of obstacles, direction. Addressing these issues will require less intrusive cues, hazard warnings.

5 Conclusion & Future Work

In this paper, we presented an AR ski exergame that integrates traditional training methods to improve turning posture and enhance motivation. We also conducted preliminary slope experiments to evaluate its feasibility and to identify remaining challenges.

Future work will focus on extending the system with adjustable training difficulty (e.g., cue speed and poses) and addressing challenges identified for deployment in ski resort environments.

Acknowledgments

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²<https://config.skytechsport.com/product/president-lux.html>

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